

Sina Hazeghi

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EDUCATION

University of Pennsylvania

Master of Science in Engineering, Robotics

- Coursework: Machine Perception, Control and Optimization, Graph Neural Networks, Intro to Mechatronics

Philadelphia, PA

Expected December 2027

Rutgers University

Bachelor of Science in Computer Science and Mathematics, Summa Cum Laude

- GPA: 3.96/4.00; SAS Academic Excellence Scholarship Recipient
- Coursework: AI, Computational/Advanced Robotics, Stochastic Processes, Theory of Statistics, Numerical Analysis

New Brunswick, NJ

December 2024

SKILLS

Languages: Java, Python, C/C++, JavaScript, Typescript, HTML, CSS, X86 Assembly, SQL, R

Technologies: AWS, CUDA, Git, JAX, Docker, GCP, Pandas, PyTorch, HuggingFace, fast.ai, Transformers, Access DB

RESEARCH

University of Pennsylvania — JIRL Lab (Loquercio Lab)

Research Assistant

- Researching egocentric human motion/pose estimation for transfer to humanoid robot control, conditioning a diffusion model to sample SMPL-H body motion from fused SLAM pose tracking and onboard sensor data
- Building on EgoAllo (Project-Aria-only spatiotemporally-invariant diffusion conditioning with test-time guidance from partially-observable hand video) and RoSHI (extends guidance to body-worn IMU sensors in place of sparse hand-frame observations); benchmarking against SAM3D as an external baseline

Philadelphia, PA

September 2026 – Present

Rutgers University — PRACSYS Lab

Student Researcher

- Evaluated differentiable physics backends (Taichi, NVIDIA Warp, JAX/Jraph) for simulating NASA's cable-driven Tensegrity robot, converging on a fully learned JAX/Jraph simulator that runs 100x faster than MuJoCo
- Trained simulators under partially-observable state data for real2sim transfer, informing research direction later formalized in "CableRobotGraphSim," accepted for oral presentation at L4DC 2026

New Brunswick, NJ

December 2023 – May 2025

SOFTWARE ENGINEERING EXPERIENCE

Bank of New York

Software Engineer

- Developed an agentic AI orchestration engine ("Eliza Holmes") using CrewAI and MongoDB to dynamically instantiate agentic workflows to accelerate client onboarding by up to 90% and winning Bank of New York's 2025 technical innovation award
- Designed a token-aware hierarchical document indexing engine with LLM summary trees and MongoDB tree-caching, cutting LLM context consumption by 70–90% and query latency from 30s to <20ms during bulk entity onboarding
- Implemented a pluggable multi-tiered guardrail framework incorporating dynamic Python validators, thread-safe execution tracking, and adversarial judge agents, catching 83% of output errors prior to end-user review

New York, NY

August 2025 – August 2026

Pfizer

Process Development Intern

- Deployed a Shiny for Python website to automate inventory processes and database integration, saving 2–3 hours/shipment

Pearl River, NY

May 2024 – August 2024

PROJECTS

Generative-Modelling | *Diffusion & Score-Based Generative Models From Scratch*

- Implemented the DiffEdit paper for text-guided image editing – generating edit masks from diffusion noise-prediction differences between two prompts, then denoising with stable diffusion
- Implemented score-based generative models in JAX, covering score matching and Langevin dynamics sampling and their connection to diffusion models

Kinodynamic-RRT | *Kinodynamic Motion Planning & Reinforcement Learning (MuJoCo)*

- Devised a Kinodynamic RRT motion planner in MuJoCo for a maze-navigation task with dynamic constraints, achieving a 96% success rate
- Implemented A2C, PPO, and SAC from scratch in PyTorch for a MuJoCo motion-generation task, with GPU-accelerated (Torch RL) distributed training reaching a 22% higher success rate than baseline